Umer Farooq
APPLIED MULTIVARIATE METHODS

Homework 06 (50 points)

Problem 1:

Dataset to be used: umaru.csv

Please do not hand in computer output separately from responses to the questions below. Instead, cut and paste relevantoutput. Copy and paste (only relevant) R codes at the end of your homework in an appendix (annotated and in chronological order).

Variable	Name	Code
Age	AGE	Years
Race	RACE	0 = White 1 = Other
Treatment	TREAT	0 = Short 1 = Long
Treatment Site	SITE	0 = A 1 = B
Returned to Drug Use Prior to Scheduled End of Program	DFREE	0 = Remained drug free 1 = Experienced relapse

Does race (RACE) modify the association between treatment site (SITE) and drug use status (DFREE)?

a) Write down the population model you will use to address this question. Be sure it is clear what notation means.(10 points)

$$logit\,(P_{DFREE}) = log\,(p/(1-p)) = \beta0 + \beta1*X_{SITE} + \beta2*X_{RACE} + \beta3*(X_{SITE}*X_{RACE}) + \epsilon$$

SITE The variable SITE represents the location of treatment, where a value of 0 denotes site A (focused on relapse prevention and health education) and a value of 1 denotes site B (dedicated to a therapeutic community program).

RACE is variable, with 0 indicating White and 1 indicating Other.

β0: is the intercept of the model indicates the log-odds of drug relapse for a white individual treated at site A.

β1 is the coefficient for the treatment site variable (SITE) signifies the difference in the log-odds of experiencing a drug relapse for a white individual at site B, relative to a white individual at site A.

β2 is the coefficient for the race variable (RACE) indicates the variation in the log-odds of a drug relapse for a non-white individual at site A, compared to a white individual at the same site.

β3 is the coefficient for the interaction term between treatment site and race (SITE×RACE) quantifies the change in the log-odds of experiencing a drug relapse for a non-white individual at site B, with all other variables held constant. This term is incorporated to assess if the relationship between the treatment site and drug use status varies based on race.

b) Fit the model and write down the fitted model. Include the appropriate output in the answer. (10 points)

Fitted model: $logit\,(\hat{P}_{DFREE}) = log\,(\hat{p}/(1-\hat{p})) = \beta0 + \beta1*X_{SITE} + \beta2*X_{RACE} + \beta3*(X_{SITE}*X_{RACE})$

$= -1.4080 + 0.5945*X_{SITE} + 0.8089*X_{RACE} - 1.2118*(X_{SITE}*X_{RACE})$

```
Call:
glm(formula = DFREE ~ SITE + RACE + SITE * RACE, family = "binomial",
    data = Umaru_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.4080     0.1478  -9.528  < 2e-16 ***
SITE          0.5945     0.2354   2.526  0.01155 *
RACE          0.8089     0.2481   3.260  0.00111 **
SITE:RACE    -1.2118     0.5071  -2.390  0.01687 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 653.73  on 574  degrees of freedom
Residual deviance: 640.82  on 571  degrees of freedom
AIC: 648.82

Number of Fisher Scoring iterations: 4
```

c) Conduct a formal test to answer the question above. To respond: write down the null and alternative hypotheses,test statistic with appropriate degrees of freedom, p-value, conclusion. Use the 5% level of significance. Include the appropriate output in the answer. (15 points)

Null Hypothesis

H0: Reduced model is sufficient and interaction term is not significant

HA: Full model is better than reduced model i.e. interaction term is significant.

Level of significance

Alpha α = 5%

Step 3: Test Statistics for the Interaction term

From the LR test which compares Full and Reduced model

From CHI2 value of 6.1 and P-Value of 0.0134 which is less than 5% and we reject the Null Hypothesis it means Full model is meaningful

```
Call:
glm(formula = DFREE ~ SITE + RACE, family = "binomial", data = Umaru_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.2988     0.1363  -9.529   <2e-16 ***
SITE          0.3055     0.2053   1.488   0.1368
RACE          0.4874     0.2124   2.295   0.0218 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 653.73  on 574  degrees of freedom
Residual deviance: 646.92  on 572  degrees of freedom
AIC: 652.92

Number of Fisher Scoring iterations: 4
```

```
> lmtest::lrtest(log_reduce_model_Umaru, log_model_Umaru)
Likelihood ratio test

Model 1: DFREE ~ SITE + RACE
Model 2: DFREE ~ SITE + RACE + SITE * RACE
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   3 -323.46
2   4 -320.41  1 6.1037    0.01349 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d) Can we interpret the "main effects" of treatment site and race in this model? Why or why not? (5 points)

In this model, the interpretation of the main effects of treatment site and race cannot be isolated from the interaction term, especially given that the interaction effect is statistically significant. This significant interaction underscores that the relationship between the treatment site and drug use status is dependent on race. Therefore, to fully understand the influence of the treatment site, one must also consider the race of the individuals involved. Likewise, the effect of race on drug use status is influenced by the treatment site, indicating that these factors do not operate independently but interact in a complex manner. Moreover, from the value of $\beta 3*(X_{SITE}*X_{RACE})$ we can also confirm the slope in full model is more pronounced.

e) Assume that this is your final model. What should you tell investigators about the relationship between treatmentsite, race, and drug use status? Summarize the results of your modeling approach in parts (b) - (d) in a short paragraph (no more than 10-12 sentences) (10 points)

1. OMNIBUS Test for the Full Model 1:
2. Likelihood test of Full and Reduce model
3. Odd Ratios for the Full Model

These three results made it clear that Interaction of Race is statistically important and provides better understanding. The likelihood ratio test assesses the significance of the interaction model by comparing the log-likelihoods of the reduced model, which excludes the interaction term, with those of the full model, which includes the interaction term. This method evaluates whether incorporating the interaction term significantly enhances the model's fit. The outcome

of this test is expressed through a Chi-squared statistic and a corresponding p-value.

Now each predictor:

For RacexSite: The significant interaction between treatment site and race, indicated by an odds ratio (OR) of 0.29767 with a p-value of 0.01687, reveals that the impact of the treatment site on the likelihood of drug relapse is influenced by the patient's race. This finding underscores the importance of tailoring treatment approaches to individual characteristics, as a universal strategy may not be as effective.

For Site: Further analysis shows that the odds ratio for the effect of the treatment site (SITE, OR = 1.81208, p = 0.01155) indicates a higher risk of relapse for participants at site B compared to those at site A. However, this risk is significantly affected by the participants' racial background, suggesting that the treatment site's impact on relapse likelihood is not uniform across different races.

For Race: Additionally, the study found a significant racial effect on drug relapse rates, with the odds ratio for race (RACE, OR = 2.24537, p = 0.00111) showing that participants of 'Other' races are at an increased risk of relapse compared to White participants, in scenarios where the interaction effect is not considered. This highlights the complexity of factors influencing drug relapse and the necessity of considering both individual and treatment-related variables in addressing substance abuse recovery.

```
setwd("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate")
library(readr)
rm(list=ls())
Umaru_data <- read.csv("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate/UMARU.csv")
install.packages("lmtest")
install.packages("car")
install.packages("MASS")
install.packages("lme4")
install.packages("glmnet")
install.packages("lmtest")
library(lmtest)
library(car)
install.packages("questionr")
summary(Umaru_data)

log_model_Umaru <- glm(DFREE ~ SITE + RACE + SITE*RACE, family = "binomial", data = Umaru_data)

summary(log_model_Umaru)

questionr::odds.ratio(log_model_Umaru)

#reduced logistic model to tests the validity/results without the interaction of race with SITE

log_reduce_model_Umaru <- glm(DFREE ~ SITE + RACE, family = "binomial", data = Umaru_data)

summary(log_reduce_model_Umaru)

questionr::odds.ratio(log_reduce_model_Umaru)

#Likelihood Ratio test for checking the significance of the interaction


lmtest::lrtest(log_reduce_model_Umaru, log_model_Umaru)
#OR and 95% CI for OR
questionr::odds.ratio(log_model_Umaru)
```